

MARINA DI GINOSA, Italy

WMO: 163250

Lat: 40.424N

Lon: 16.887E

Elev: 12

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.0	(c) 2.3	(d) -4.0	(e) 2.7	(f) 8.2	(g) -2.0	(h) 3.2	(i) 7.0	(j) 14.8	(k) 13.4	(l) 13.4	(m) 13.4	(n) 4.6	(o) 290	(p) 0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 34.8	(d) 23.6	(e) 33.0	(f) 23.8	(g) 31.8	(h) 23.7	(i) 28.3	(j) 31.0	(k) 27.2	(l) 30.3	(m) 26.2	(n) 29.5	(o) 4.7	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.6	23.5	30.6	26.2	21.7	29.6	25.2	20.3	28.8	91.4	30.9	86.3	30.4	81.7	29.8	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	9.9	8.3	-0.9	37.9	1.2	2.2	-1.8	39.5	-2.5	40.8	-3.2	42.1	-4.0	43.7
			-1.8	29.9	1.2	1.9	-2.6	31.3	-3.3	32.4	-4.0	33.4	-4.9	34.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	9.4	9.7	11.9	14.5	19.0	23.4	26.5	26.5	22.4	18.4	14.0	10.5
	DBStd	6.69	2.70	2.76	2.65	2.56	2.57	2.84	2.29	2.30	2.63	2.70	3.08	2.87
	HDD10.0	125	43	34	13	2	0	0	0	0	0	0	6	27
	HDD18.3	1272	278	241	201	118	22	1	0	0	3	32	133	242
	CDD10.0	2762	24	27	70	137	280	403	512	510	373	259	125	43
	CDD18.3	865	0	0	0	2	43	154	254	252	126	33	2	0
	CDH23.3	8115	0	0	2	10	246	1334	2826	2721	884	91	2	0
	CDH26.7	3072	0	0	0	1	42	445	1232	1149	199	4	0	0
Wind	WSAvg	4.0	4.2	4.4	4.4	4.0	3.8	3.7	3.8	3.7	3.8	3.8	4.1	4.0
Precipitation	PrecAvg	449	50	43	42	30	28	21	17	20	42	57	58	58
	PrecMax	720	212	124	130	156	84	78	77	111	183	197	205	239
	PrecMin	170	0	1	1	2	1	0	0	0	0	0	0	0
	PrecStd	150	49	33	30	28	20	19	19	22	43	47	49	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	18.6	21.9	24.5	30.0	35.2	37.1	37.0	32.1	27.2	23.0	19.1
		MCWB	13.6	13.3	13.1	16.6	20.2	22.6	22.1	23.9	22.8	21.0	19.5	15.8
	2%	DB	16.1	16.9	19.5	22.2	27.8	32.3	34.9	34.8	30.2	25.2	21.2	17.6
		MCWB	13.6	13.3	13.4	15.6	19.6	21.7	23.6	24.1	23.5	20.8	18.2	14.8
	5%	DB	15.0	15.8	17.9	20.9	26.0	30.9	33.2	33.0	28.9	24.1	20.1	16.6
		MCWB	12.9	12.9	13.5	15.2	19.4	22.3	23.7	24.6	23.5	20.3	17.4	14.1
	10%	DB	14.1	14.9	16.3	19.2	24.2	29.2	32.0	31.5	27.7	23.1	19.1	15.2
		MCWB	12.1	12.5	13.2	15.1	19.1	22.2	23.8	24.5	23.0	19.8	16.8	13.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.7	16.3	19.2	23.7	27.7	29.6	29.7	27.3	23.7	20.4	16.8
		MCDB	16.8	16.9	18.3	22.1	26.5	30.0	32.4	32.3	29.3	25.4	21.9	18.3
	2%	WB	14.3	14.7	15.3	17.7	22.2	26.0	28.3	28.5	26.1	22.3	19.2	15.5
		MCDB	15.7	15.9	17.3	20.3	25.3	29.0	31.1	31.3	28.3	24.0	20.5	17.0
	5%	WB	13.4	13.8	14.5	16.7	21.1	24.9	27.2	27.5	24.9	21.3	18.0	14.3
		MCDB	14.8	15.3	16.6	19.1	24.2	28.4	30.5	30.4	27.5	23.1	19.7	15.9
	10%	WB	12.3	12.6	13.7	15.9	20.1	23.9	26.1	26.5	23.7	20.4	17.0	13.4
		MCDB	14.0	14.4	15.9	18.1	23.0	27.3	29.8	29.7	26.5	22.5	19.0	15.1
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.0	8.2	8.6	8.9	9.3	9.8	9.6	8.9	8.2	7.5	7.3
		MCDBR	8.4	9.4	10.6	11.0	11.2	11.4	11.7	11.1	9.9	9.1	8.2	7.9
		MCWBR	6.5	6.8	6.8	6.7	6.6	6.7	7.3	7.0	6.7	6.3	6.3	6.1
	5% WB	MCDBR	7.4	8.0	8.7	9.3	9.5	9.6	9.9	9.4	8.7	7.9	6.7	7.0
		MCWBR	6.5	6.9	6.8	7.0	7.1	7.4	8.2	8.0	7.4	6.5	5.9	6.1
Clear-Sky Solar Irradiance	taub	0.343	0.366	0.405	0.453	0.449	0.435	0.430	0.438	0.432	0.401	0.371	0.347	
	taud	2.447	2.358	2.248	2.120	2.195	2.280	2.299	2.281	2.293	2.364	2.398	2.437	
	Ebn at Noon	811	840	842	823	835	845	845	827	806	790	765	772	
	Edh at Noon	80	101	125	152	144	132	129	128	118	99	84	76	
All-Sky Solar Radiation	RadAvg	1.89	2.64	3.92	5.09	6.21	7.02	7.24	6.38	4.66	3.27	2.14	1.70	
	RadStd	0.19	0.23	0.40	0.41	0.37	0.38	0.32	0.38	0.26	0.27	0.18	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	-61	+91	N/A

Nomenclature: See separate page